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SUPERSEDING
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FEDERAL SPECIFICATION

SHOWER HEAD, BALL JOINT (INTEGRAL FLOW CONTROL)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metallic and plastic shower heads with a self-contained flow control.

1.2 Classification. Shower heads shall be of the following types, as specified (see 6.2):

Type I - Metallic.
Type II - Nonmetallic.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Publications:

B 287 - Acetic Acid-Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Standard commercial product. Shower heads covered by this specification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's current standard commercial product.

3.2 Materials.

3.2.1 Metals. Metallic materials shall be brass or stainless steel.

3.2.2 Nonmetallic. Nonmetallic materials shall be at the manufacturer's option. The materials shall be suitable for the intended purpose and of the quality used in the manufacture of his standard commercial product.

3.3 Construction.

3.3.1 General. The shower head shall consist of the ball joint assembly, shower head shell, and face plate. Washers, gaskets, and snap rings used in sealing, connecting or securing one section to another may be metallic or nonmetallic.

3.3.1.1 Type I - metallic shower heads. The type I shower head body shall be all metal construction except for the face plate which shall be metal, plastic, or a combination thereof.

3.3.1.2 Type II - nonmetallic shower heads. The type II shower head shall be a shower head conforming to this specification, having a plastic shell or a plastic ball joint.

3.3.2 Inlet connection. The inlet connection shall be internally threaded with 1/2 inch iron pipe size (IPS) threads.

3.3.3 Flow controls. Flow controls shall automatically regulate the flow rate (see 3.6).

3.3.3.1 Type I shower head. Flow controls for type I shower heads shall be integral with the body, shell, or ball joint assembly, or shall be contained within the shell.

3.3.3.2 Type II shower head. Flow controls for type II shower heads shall be at the manufacturer's option.

3.3.4 Cleansing. The shower head face plate shall permit the easy removal of mineral deposits, rust, scale, or other impurities without the use of special tools. When the disassembly of a shower head is required for cleansing, instructions for the disassembly and reassembly shall be included in each shower head package.

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3.4 Finish.

3.4.1 Metals.

3.4.1.1 Stainless steel. All visible exterior stainless steel surfaces shall have a number 4 general purpose polish finish produced with a 120-150 mesh abrasive or equal.

3.4.1.2 Brass. Exterior brass surfaces shall have a bright chromium plate finish. The plated surface shall show no evidence of corrosion when tested for 8 hours in accordance with 4.4.1.1.

3.4.2 Plastic. Plastic shower heads shall have an as-molded or chromium plated finish. The color of unplated exposed plastics shall be black, gray or white throughout. Chromium plated shower heads shall have not more than 4 void areas not greater than 1/32 inch in diameter when examined in accordance with 4.3.1.1. Plating shall adhere to the shower head when tested in accordance with 4.4.1.2.

3.5 Water cone spread. The water cone spread shall be an even conical spray measuring 2-1/2 ± 1 foot in a 6-1/2 foot vertical drop (see 4.4.3).

3.6 Volume regulator flow rate. The rate of water flow through the shower head shall not be greater than 3.0 GPM when tested in accordance with 4.4.4.

3.7 Marking. Each shower head shall be permanently and legibly marked with manufacturer's readily identified registered trademark or name.

3.8 Workmanship. All shower heads shall be free of crazing, seam lines, burrs, sharp edges, and surface nicks and pits. All exterior surfaces shall be uniform in appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105, inspection level S-2.

4.3 Examination and tests.

4.3.1 Examination before testing. Samples selected in accordance with 4.2 shall be examined for the defects listed in table I. The acceptable quality levels (AQL's) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Class not as specified (see 1.2)	X	
Shower head not cleanable (see 3.2)	X	
Threads not as specified	X	
Finish on stainless steel not as specified (see 3.4)		X
Marking not as specified		X
Crazing, seams, burrs, surface nicks and pits, or sharp edges on shower heads		X
Voids on chromium plated plastic surfaces		X
Surfaces not of uniform appearance		X

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4.3.1.1 Void areas. Examine shower heads under a hooded 100 watt lamp at a distance of 18 inches ± 6 inches for conformance to 3.4.2.

4.3.2 Tests. Shower heads selected in accordance with 4.2 shall be tested as specified in 4.4. Unless otherwise specified, the AQL for tests shall be 4.0 percent defective.

4.4 Test methods.

4.4.1 Finish.

4.4.1.1 Chromium plating on metal. Test plating on metal in accordance with ASTM B287. At end of test, examine shower heads for conformance to 3.4.1.2.

4.4.1.2 Adhesion of chromium plating on plastics. Test chromium on plastics for conformance to 3.4.2 as follows:

(a) Place the units in an oven heated to 160° ± 5°F. Leave them in the oven for a period of at least 1 hour. Remove from oven and air cool to room temperature.

(b) Allow units to remain at room temperature for 1 hour ± 5 minutes.

(c) Place the units in a cold chamber maintained at -20° ± 2°F. Leave them in the chamber for a period of at least 1 hour. Remove from the chamber and air warm to room temperature.

(d) Allow units to remain at room temperature as in (b) above.

(e) Scribe an X of convenient size through the coat using a razor blade or sharp knife.

(f) Place a piece of 3/4 inch wide transparent tape over the X and press firmly against the X, then remove quickly.

4.4.2 Hot water conditioning. Flow water at 140 ° ± 5°F. for one hour through shower heads prior to testing in accordance with 4.4.3 and 4.4.4. In lieu of this conditioning, shower heads tested in accordance with 4.4.1.2 shall be used for tests specified in 4.4.3 and 4.4.4.

4.4.3 Water cone spread. Mount shower head to a 1/2 inch supply line. Maintain a flowing water pressure of 25 psig in the line at the shower head for the duration of the test. Observe and measure the spray cone for conformance to 3.5.

4.4.4 Flow rate. Connect shower head to a smooth pipe or tube of the same nominal diameter as the shower head inlet. The minimum length of the pipe or tube shall be 20 inside diameters when measured from the ball joint. A pressure gauge shall be located 1/2 to 2-1/2 diameters upstream from the shower head. Allow water to run through pipe or tube and shower head at a flowing pressure of 80 psig. Catch water in a suitable container for a one minute period. Repeat test to complete 3 cycles. Average results to obtain gallons per minute. Compare results for conformance to 3.6. Failure of any sample to pass this test shall be cause for rejection.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine whether the packaging, packing and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operation. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted, incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing, damaged or not conforming to the applicable specification or requirements stated herein.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Quantity per container is more or less than required. Gross weight exceeds requirements of the box specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each shower head shall be packaged in a paperboard or fiberboard box. Twelve packaged shower heads shall be further packaged in a paperboard or fiberboard box. The boxes shall be secured to prevent accidental opening.

5.1.2 Commercial. Each shower head shall be packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.2.1 Level A. Twelve intermediate packages (144 shower heads) packaged as specified in 5.1.1, shall be packed in a box made from weather resistant fiberboard having a bursting strength of not less than 275 pounds per square inch. The box flaps shall be secured with water resistant adhesive applied to not less than 75 percent of the surface area of contact between the flaps or with 3-inch wide water resistant tape applied to the full length of the center seams and extending over the ends not less than 3 inches. Alternatively, cleated plywood, wirebound, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The edges of the barrier material shall be sealed with waterproof tape or adhesive.

5.2.2 Commercial. The shower heads packaged as specified in 5.1.2 shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commerical practice. The unitized load shall not exceed 2,500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Packages, shipping containers, and unitized loads (when applicable) shall be marked in accordance wtih Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Shower heads covered by this specification are intended for residential and other similar interior use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Specification part number (see 1.2.1).
(c) Level of packaging and packing required (see 5.1 and 5.2).

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6.3 Shower heads covered by this specification meet all the requirements for nonadjustable spray water saver shower heads in accordance with Federal Specification WW-P-541.

MILITARY INTERESTS:

Custodians:

Navy - YD
Air Force - 99

Users Activities:

Navy - MC
Air Force - 99

Preparing activity:

GSA - FSS

Civil Agency Coordinating Activities:

GSA - PCD, FSS
TVA - TVA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

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